

Lab 0 - Rubber Stoppers

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February 12, 2026

1 Purpose

The purpose of this lab is to determine the density of two sizes of rubber stoppers and to write up a complete lab report using proper format.

2 Introduction

Density is a physical property that describes the relationship between the mass of a material and its volume. This property may be used to identify a material and can be used to compare two different materials. Density is the ratio calculated using $\rho = \frac{m}{V}$, where ρ is the density, m is the mass, and V is

Water-displacement and mass measurements were used to determine the volumes and densities of two rubber stoppers. The smaller stopper volume was determined by graduated-cylinder displacement; the larger stopper volume was determined by measuring the change in mass of a watch glass after overflow collection. Calculations were performed using measured masses and displaced-volume values to obtain densities.

Glassware was handled with care to avoid breakage or spills. Broken glass was disposed of by qualified personnel.

3 Materials

- Graduated cylinder (200 mL)
- Two rubber stoppers (small & large)
- Beaker (50 mL)
- Watch glass
- Water
- Mass scale
- Dropper

4 Procedure

4.1 Part 1: Density of Small Rubber Stopper

Materials/Equipment

Actions Taken

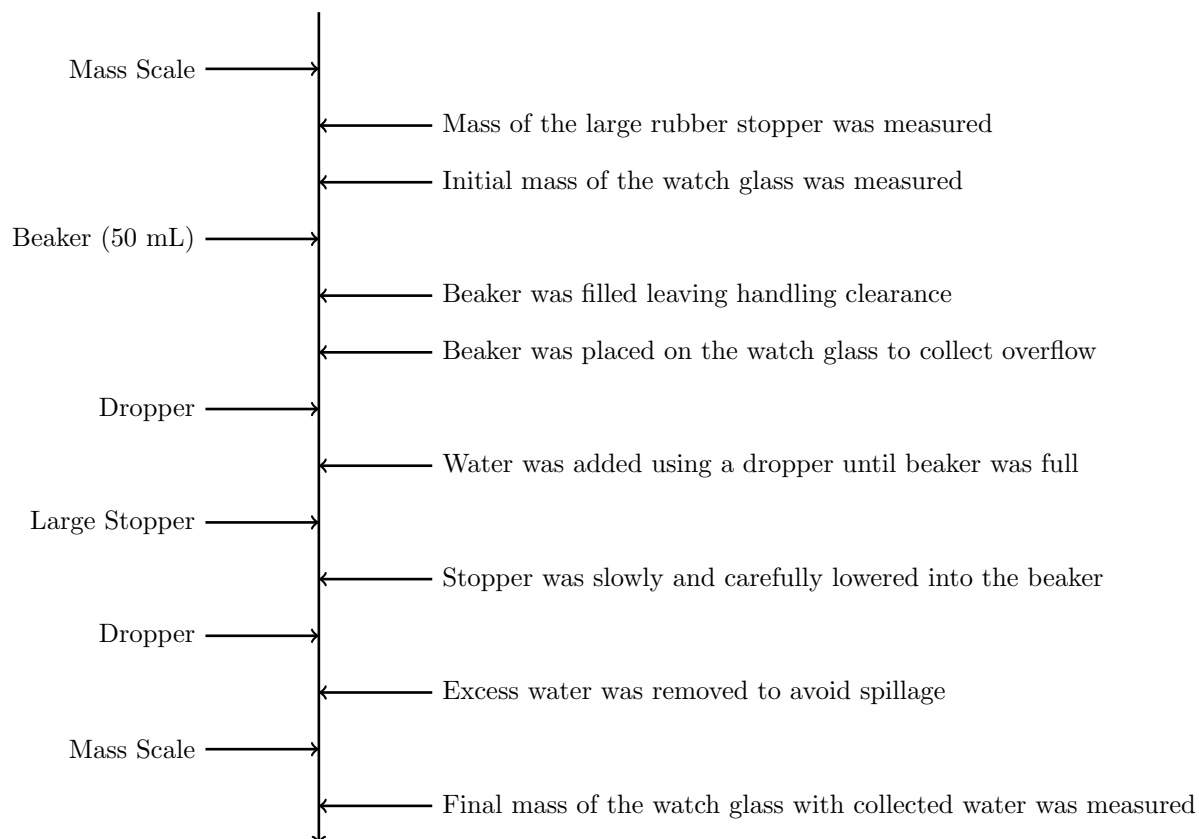


Mass Scale	→	←	Mass of the small rubber stopper was measured
Graduated Cylinder	→	←	Graduated cylinder was filled to an intermediate level
		←	Initial water level was recorded
Small stopper	→	←	Small rubber stopper was lowered into the cylinder
		←	Water was stirred to remove trapped bubbles
		←	Final water level was recorded

4.2 Part 2: Density of Large Rubber Stopper

Materials/Equipment

Actions Taken



5 Observations

Table 1: Observations of Rubber Stoppers

Rubber Stopper Size	Mass (g)	Initial Measurement	Final Measurement
Small	4.33g	15.1 mL	16.8 mL
Large	24.25g	25.59g	41.07g

6 Analysis

6.1 Density of Small Rubber Stopper

$$\begin{aligned}\text{Volume} &= \text{Final} - \text{Initial} \\ &= 16.8 \text{ mL} - 15.1 \text{ mL} \\ &= 1.7 \text{ mL} \\ &= 1.7 \text{ cm}^3\end{aligned}$$

$$\begin{aligned}\text{Density} &= \frac{\text{Mass}}{\text{Volume}} \\ &= \frac{4.33 \text{ g}}{1.7 \text{ cm}^3} \\ &\approx 2.55 \text{ g/cm}^3 \\ &\approx 2.6 \text{ g/cm}^3\end{aligned}$$

$\therefore$ the density of the small rubber stopper was approximately 2.6 g/cm^3 .

6.2 Density of Large Rubber Stopper

Assuming the density of water is 1 g/mL .

$$\begin{aligned}\text{Volume} &= (\text{Final Mass} - \text{Initial Mass}) \times 1 \text{ g/mL} \\ &= 41.07 \text{ g} - 25.59 \text{ g} \\ &= 15.48 \text{ g} \\ &= 15.48 \text{ mL} \\ &= 15.48 \text{ cm}^3\end{aligned}$$

$$\begin{aligned}\text{Density} &= \frac{\text{Mass}}{\text{Volume}} \\ &= \frac{24.25 \text{ g}}{15.48 \text{ cm}^3} \\ &\approx 1.567 \text{ g/cm}^3\end{aligned}$$

$\therefore$ the density of the large rubber stopper was approximately 1.567 g/cm^3 .

7 Discussion

1. Using value from Analysis of Part 1 and Part 2:

$$\begin{aligned}\text{Percentage Error} &= \left| \frac{\text{Experimental Value} - \text{Theoretical Value}}{\text{Theoretical Value}} \right| \times 100\% \\ &= \left| \frac{1.567 - 2.55}{2.55} \right| \times 100\% \\ &\approx 38.5\%\end{aligned}$$

$\therefore$ the percentage error for the method from Part 2 was approximately 38.5%.

2. To find the percentage composition of copper and zinc in a brass sample:

- (a) Take the mass of the brass sample using a mass scale.
- (b) The volume of the brass sample would be determined using water displacement in a graduated cylinder.
- (c) The density would be calculated using the formula $\rho = \frac{m}{V}$.
- (d) The percentage composition of copper and zinc in the brass sample could be determined by using a system of equations with the known densities of pure copper and pure zinc.

8 Conclusion

The density of the small rubber stopper was approximately 2.6 g/cm^3 , and the density of the large rubber stopper was approximately 1.567 g/cm^3 . The percentage error for the method from Part 2 was approximately 38.5%. A full lab report was written with a detailed procedure flowchart, observations table, analysis, and discussion of results.